untitled (3.13.17)

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# TASK 10: Compare Expectations with Execution

BASE_RATE = 30.00
FAT_STEP_VALUE = 0.40
SNF_STEP_VALUE = 0.30

REFERENCE_FAT = 3.5
REFERENCE_SNF = 8.5

MIN_FAT = 3.0
MAX_FAT = 6.0


def calculate_milk_rate(fat, snf):
    if fat < MIN_FAT or fat > MAX_FAT:
        raise ValueError("Invalid fat value")

    if snf <= 0:
        raise ValueError("Invalid SNF value")

    fat_steps = round((fat - REFERENCE_FAT) * 10)
    snf_steps = round((snf - REFERENCE_SNF) * 10)

    rate = BASE_RATE + fat_steps * FAT_STEP_VALUE
    rate += snf_steps * SNF_STEP_VALUE

    return round(rate, 2)


def calculate_payment(litres, fat, snf):
    if litres <= 0:
        raise ValueError("Invalid milk quantity")

    return round(litres * calculate_milk_rate(fat, snf), 2)


def run_test(name, litres, fat, snf, expected):
    try:
        actual = calculate_payment(litres, fat, snf)
    except ValueError:
        actual = "Invalid input"

    if actual == expected:
        result = "PASS"
    else:
        result = "FAIL"
```

```
        actual = "Invalid input"

    if actual == expected:
        result = "PASS"
    else:
        result = "FAIL"

    print(name)
    print("Expected:", expected)
    print("Actual:", actual)
    print("Result:", result)
    print("----------------------------")


print("MILK PAYMENT CALCULATOR - TEST RESULTS")
print()

run_test("Test 1", 10, 3.5, 8.5, 300.0)
run_test("Test 2", 10, 4.0, 8.7, 326.0)
run_test("Test 3", 10, 3.0, 8.0, 275.0)
run_test("Test 4", -5, 3.5, 8.5, "Invalid input")
run_test("Test 5", 10, 7.0, 8.5, "Invalid input")
```